MK vision
Smart Secure Style

Catalogue

PRODUCTS & SOLUTIONS

MK Vision Joint Stock Company

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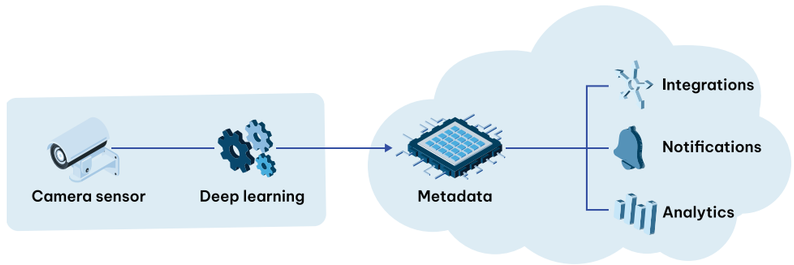
1 MK VISION INTRODUCTION

MK Vision Joint Stock Company, founded in 2020, is a manufacturer and provider of high-tech camera products and AI-powered security surveillance solutions in Vietnam. The company is a member of MK Group, a technology corporation with over 25 years of experience in the information technology sector. MK Group has successfully implemented numerous large-scale projects for corporate and government clients and is proud to be a strategic partner of the Research and Application Center for Residents Data (RAR Center) under Vietnam's Ministry of Public Security. Over the years, MK Vision's technological expertise and innovative solutions have earned recognition from customers and partners both locally and globally.

With the spirit of "Resilient Production - Technological Autonomy - Driving Innovation," MK Vision's products are the culmination of rigorous research and development by the company's engineers and are proudly manufactured entirely in Vietnam. We are unwavering in our commitment to innovation, striving to deliver cutting-edge products that empower customers to live safer lives through modern technology.

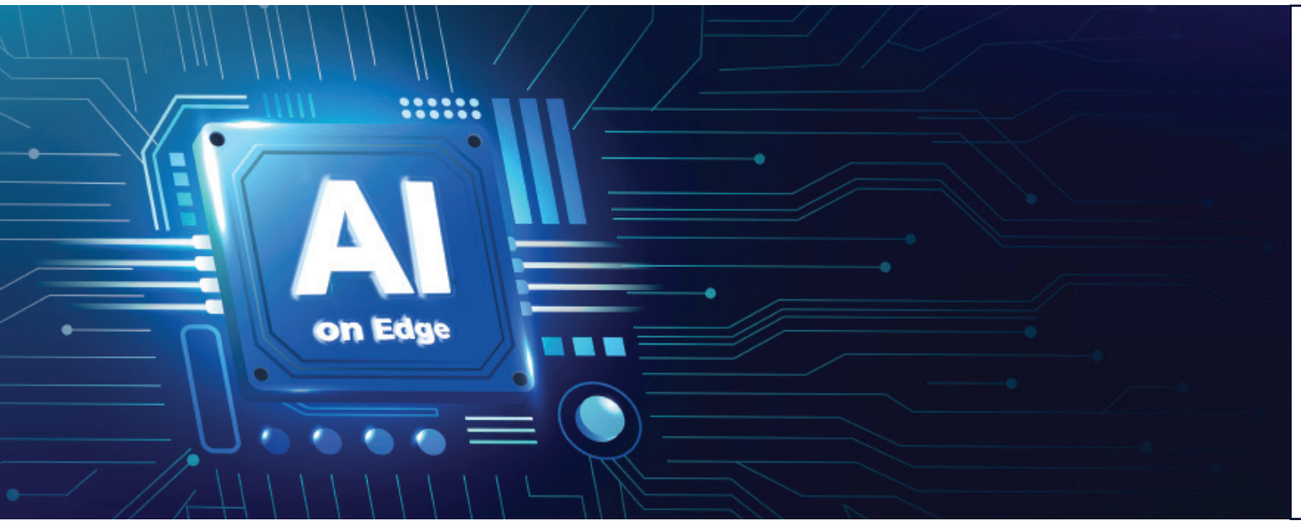
Image: MK Vision Factory

AI ON EDGE TECHNOLOGY



The AI on Edge technology facilitates faster data processing, enhances data security, reduces network congestion, and enables offline operation when an Internet connection is unavailable by executing algorithms locally on edge devices.

AI on Edge introduces cost-effective, scalable, and secure deployment models for cameras, enhancing data privacy. The future of AI on Edge promises groundbreaking advancements, fostering critical innovations and driving growth across various industries, particularly in the field of security surveillance.



The Benefits of Edge AI

Bandwidth Efficiency

AI on Edge minimizes bandwidth usage by processing data locally on the device, thereby reducing the volume of data that needs to be transmitted to the cloud for analysis. This optimization of network bandwidth usage lowers data transmission costs and reduces network congestion, particularly in environments with limited available bandwidth.

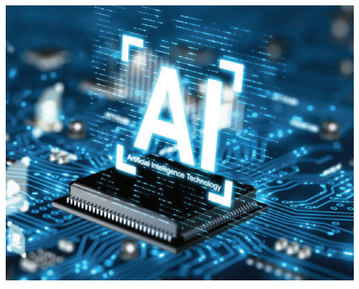
Real-time Performance

Edge AI significantly reduces latency by processing data locally on devices. Unlike cloud-based processing, edge processing completes tasks in just a few milliseconds. This enables real-time decision-making and rapid responses, making Edge AI ideal for applications demanding instant results, such as industrial automation and quick-response security operations.

Enhanced Data Security

Edge AI also enhances data security by processing sensitive information locally on devices, reducing the risk of data leakage or cyberattacks during transmission to cloud servers. This approach ensures that critical data remains confined to the device, minimizing the threats of unauthorized access, man-in-the-middle attacks, or eavesdropping.

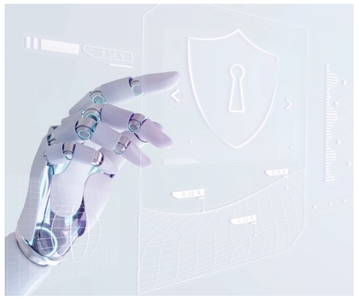
MK Vision's Technological Commitment



Providing customers with high-quality cameras at affordable prices, meeting the most stringent demands for information security.

Leading the adoption of cutting-edge technologies, ensuring compliance with government regulations, and offering versatile applications across various aspects of life, becoming an indispensable tool for organizations and businesses.

Reasons to Choose MK Vision



Technological Autonomy

MK Vision takes pride in being a leader in the security surveillance industry, leveraging cutting-edge technology through technological autonomy in the research, development, and manufacturing of AI cameras in Vietnam. We own a factory in Vinh Phuc that produces ISO 9001:2015-certified cameras, supported by a product development team composed of leading experts in optics, imaging, and data processing.

MK Vision is a Trusted Partner

TRUNG TÂM NGHIÊN CỨU, ỨNG DỤNG DỮ LIỆU DÂN CƯ VÀ CĂN CƯỚC CÔNG DÂN
RESEARCH AND APPLICATION CENTER FOR RESIDENTS DATA (RAR)

MK Vision is a member of MK Group, a technology corporation with over 25 years of experience in the information technology sector. MK Group has successfully implemented numerous large-scale projects for corporate and government clients and is proud to be a strategic partner of the RAR Center under Vietnam's Ministry of Public Security. Over the years, MK Vision's technological expertise and innovative solutions have earned recognition from customers and partners both locally and globally.

Pre- and Post-Sales Support

With a nationwide network of dealers and partners, MK Vision is committed to provide fast and reliable 24/7 support, warranty, and maintenance services. We are always ready to ensure the uninterrupted operation of systems and to address any issues promptly and effectively. Additionally, with our product research and development capabilities, MK Vision is equipped to deliver customized solutions and meet specific product development requirements for individual clients and specialized operations.

MK Vision at Sao Khue Award 2024

MK Vision at Vietnam International Innovation Exhibition 2023 (VIIE 2023)

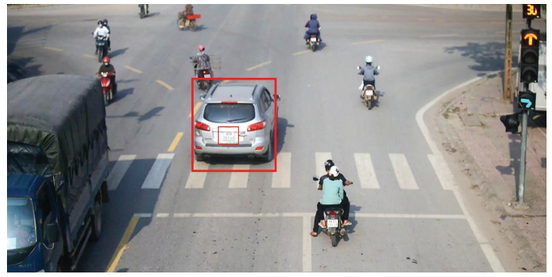
2 DEMAND OF AI CAMERAS

MK vision

Government Sector Solutions

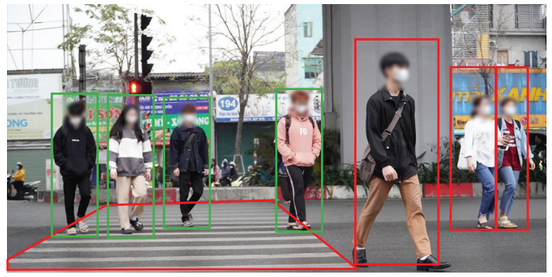
Traffic Monitoring and Management

AI-powered cameras integrated into Intelligent Transportation Systems (ITS) significantly enhance traffic management and operations, reduce congestion and accidents on roads, while promoting greater public compliance with traffic safety regulations.



Urban Security

AI cameras, integrated with urban security management and monitoring solutions, are employed for tasks such as object detection, crowd monitoring, identifying incidents involving fights or weapon use, and preventing sidewalk and roadway encroachments.



Police

AI cameras support real-time monitoring to collect evidence, aiding in investigations and enforcing penalties for individuals violating the law.

Border Security

AI cameras are designed to operate in the harshest conditions, assisting border forces in tracking and apprehending individuals involved in illegal immigration or border regulation violations. These systems contribute to maintaining social safety, ensuring territorial sovereignty, and strengthening national security.

Suggested Products

V-ANPR

Surveillance camera detecting traffic violations

V-Security

Security surveillance camera

V-SPEED

Speed measurement camera with recording capability

AutoSecure

Smart dash camera

BWC

Body-worn operational camera

V-PTZ

360-degree security surveillance camera

Government Sector Solutions

Aviation - Immigration

High-tech security solutions are applied to enhance the efficiency of management and monitoring at airports and aviation hubs. These solutions help minimize potential incidents, streamline immigration management, and oversee staff activities effectively.

Environmental Monitoring

AI Cameras are utilized to monitor and oversee water, forest, and urban environments, detecting violations such as illegal waste discharge and issuing warnings about potential forest fires. These technologies play a vital role in raising community awareness about environmental protection.

Education

Monitoring and attendance devices integrating biometric technology and smart cards at educational institutions enhance security and improve accountability in managing student numbers.

Healthcare

AI cameras and security monitoring devices are used to manage the flow of visitors and workers in hospital areas, monitor, and warn of safety-security risks occurring in hospitals.

Suggested Products

V-Surveillance

Surveillance IP camera

V-Security

Security surveillance camera

AutoSecure

Smart dash camera

M6

E-ID and biometric reader

BWC

Body-worn operational camera

V-PTZ

360-degree security surveillance camera

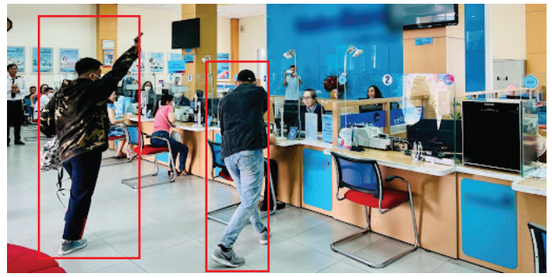
Enterprise Sector Government Sector Solutions

Retail

AI cameras help monitor store activities, analyze customer shopping behavior to optimize experience and inventory management, detect and prevent theft, ensuring safety for staff and customers.

Banking

AI Cameras integrated with enhanced security monitoring systems for banks to help alert security risks, including customer impersonation and theft.



Suggested Products

V-Security

Security surveillance camera

BWC

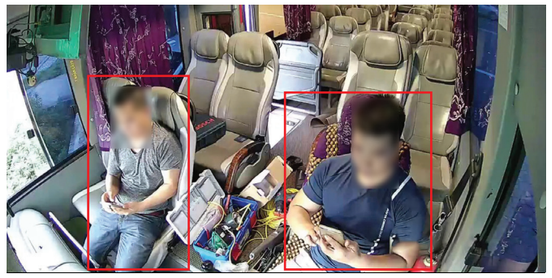
Body-worn operational camera

V-PTZ

360-degree security surveillance camera

Transportation

AI Cameras are used as per Decree 10/2020/ND-CP for dedicated monitoring of transport companies, supporting fleet management and driver behavior, enhancing efficiency, and ensuring driver safety.

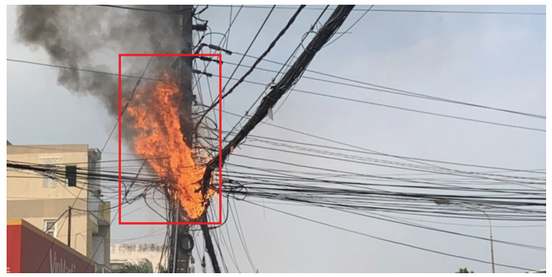


Construction Sites

AI Cameras monitor construction progress, ensure worker safety, and track valuable materials on site, reducing theft risk.

Electricity

AI cameras monitor power stations, analyzing data to detect abnormal activities, prevent incidents, and reduce risks. The technology supports infrastructure management, ensuring employee safety, and protecting critical equipment.



V-Surveillance

Surveillance IP camera

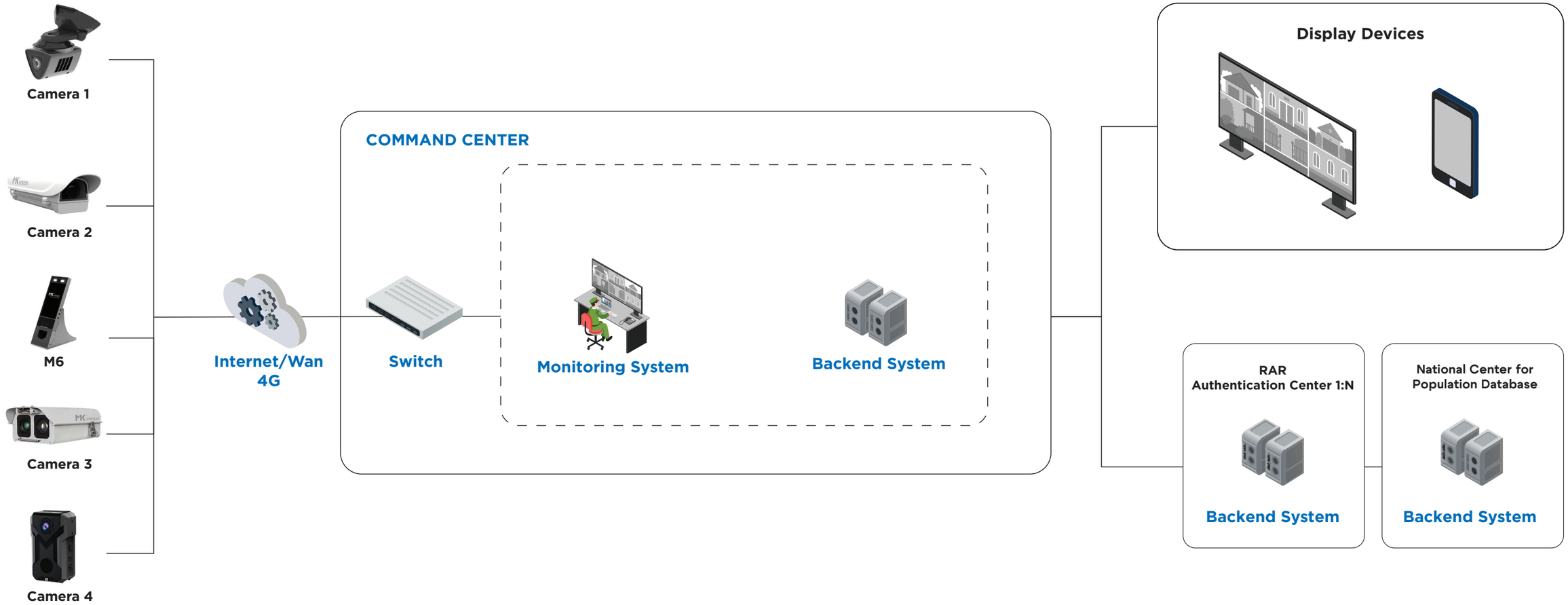
AutoSecure

Smart dash camera

3 PRODUCTS - SOLUTIONS

MK vision

Overall Connection Model



Dashcam Solution

Smart dash camera

AutoSecure

AutoSecure - the first intelligent dashcam in Vietnam integrated with AI to monitor driver behavior. With 4K resolution and a Driver Monitoring System (DMS), AutoSecure provides a comprehensive solution that gives fleet managers a multidimensional perspective on their company's transportation activities. The product complies with Decree 10 and Circular 12 of the Ministry of Transport.

- **High-Speed Connectivity**
  Real-Time Reporting
- **Precision Acceleration Sensor**
  Customized to match users' demand
- **AI Image Processing Chip**
- **Super energy - saving**
- **Intrusion Reports**
- **Driver's Face Recognition**
- **Remote Video Monitoring**
  View from the front of the vehicle and inside the cabin
- **Driver Monitoring System**
  Supervise 7 driver behaviors

Intelligent Dash Cam

AutoSecure is equipped with automatic recording capabilities combined with accelerometers to automatically detect collisions, non-compliant driving, or accidents. The camera will automatically record the driving incident and lock the video to serve as critical evidence for insurance and investigation purposes.

Smart Vehicle Protection

AutoSecure is equipped with various technologies to support safe driving and meet advanced management needs, including:
The FleetFocus function utilizes accelerometers to support automatic video locking and protection when sudden changes in velocity are detected, such as sudden braking, sharp turns, acceleration, and collisions. Furthermore, with the FleetFocus+ upgrade package, the camera can also recognize dangerous driving behaviors such as drowsiness, smoking, fatigue, lack of alertness, distraction, using a phone, and not wearing a seatbelt, employing artificial intelligence (AI) technology.

Real-Time Management

Integrated with MK Vision's software, AutoSecure provides the capability to directly monitor video from the camera, recording all notifications in real time. Additionally, the camera incorporates GPS positioning and fleet management features, enabling the management center to easily capture and promptly handle emerging situations.

Automated Examination and Assessment Solution

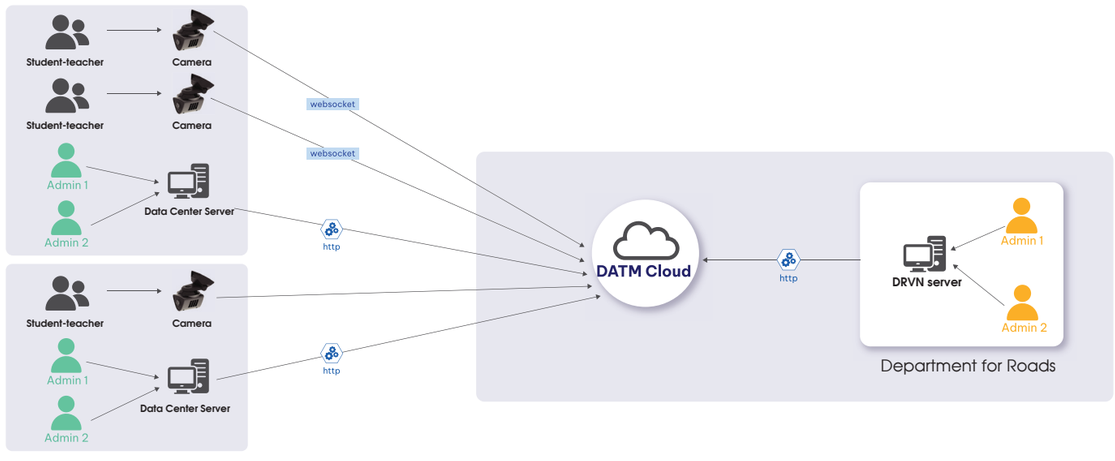
Autosecure DAT

AutoSecure DAT is a Distance and Time (DAT) monitoring device installed on automobiles to record, store, authenticate, and transmit essential information related to driver training and practice sessions. According to the regulations set by the Ministry of Transport, the use of DAT devices is mandatory during driving practice to ensure the accuracy and objectivity of evaluation results. AutoSecure DAT is designed to comply with QCVN 105:2020/BGTVT standards and includes enhanced features such as: identification of students/instructors via ID cards, full-journey monitoring, specialized face recognition cameras (anti-glare and anti-backlight).

Benefits

- Information display and operational status notifications via lights, sound, or screen, including: mobile network signal, server connection status, GNSS signal, storage status, camera operation status, instructor/student login/logout status, real-time driving practice time and distance, and remaining time requirements as per regulations.
- Functionality to record changes of instructors and students on the vehicle using ID cards according to protocols by the Ministry of Public Security.
- Real-time student authentication via specialized cameras (anti-glare and anti-backlight).
- Data recording and storage capabilities for the instructor and student information, vehicle journey details, and student face authentication results, with data transmission to the management server within the specified time frame.

DAT connection model



Body worn camera solution

Body-worn operational camera

Body Worn Camera

BODYWORN is a mobile device that can be worn on the body, providing the capability to record real-life situations during the work processes of officers and employees.

Especially in emergency situations where quick response and directives from the operation center are required, it helps ensure that all activities are accurately monitored, supported, and managed.

AI PROCESSING CHIP
Equipped with a dedicated NPU (Neural Processing Unit)

OPERATIONAL TIME
12+ hours (per charging time)

DUST AND WATER RESISTANCE INDEX
IP 65

NFC COMMUNICATION
Protocol supports identity verification via embedded-chip e-ID and electronic-passport

GPS
GPS & GLONASS / A-GPS

VIDEO
Resolution: 1080p

SUBJECT SEARCHING
Face Recognition

WIRELESS CONNECTIVITY
4G LTE

WEIGHT
205G

First AI Body Camera in Vietnam

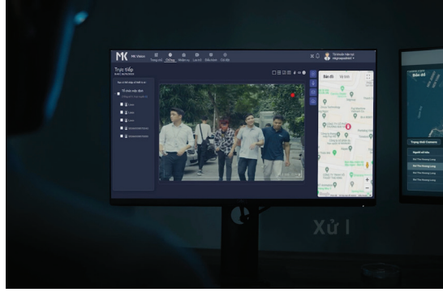
Utilizing AI technology, the device can perform facial recognition to detect and identify wanted persons or missing individuals, aiding the police in investigations and criminal apprehension. Facial data can be verified against the Ministry of Public Security's citizen database.

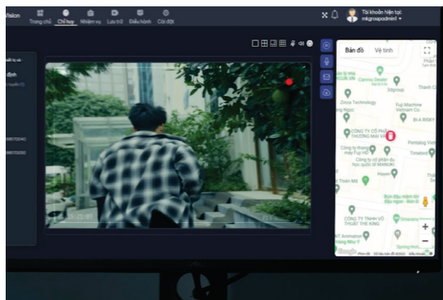
ID Card Reading Capability

The BWC is equipped with NFC-based ID card reading, saving time during identity checks in patrols and searches. This essential feature is developed by MK Vision in collaboration with the RAR Center (C06) for market introduction.

Command Center Software

Management software provides statistics on the status of devices and stores patrol data, showing the location of each device to aid in supervision, report extraction, evidence data management, and recognition list maintenance. Commanders can also view live video and issue orders to teams, facilitating remote monitoring and timely situation control.





Access Control Devices Using ID Cards

E-ID and biometric reader

M6

MK Vision's M6 Biometric Identity Card Authentication Device is a card reader for identity cards using fingerprint/face matching (MOC) protocols as defined by the Ministry of Public Security. It can connect with access control software developed by MK Vision.

This product supports businesses and organizations in enhancing security, verifying identities, and managing access across various sectors such as e-government, smart cities, online payment banking, healthcare, transportation, education, and telecommunications.

Fingerprint
FAP 20 optical

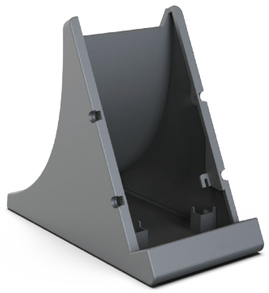
Wifi
Wifi 802.11 b/g/n (2.4GHz)

Integrated Lighting
White LED

NFC protocol
Supports reading chip-embedded e-ID and e-passport

Wired connection
LAN (PoE)

Screen
5-inch touchscreen



Access Control

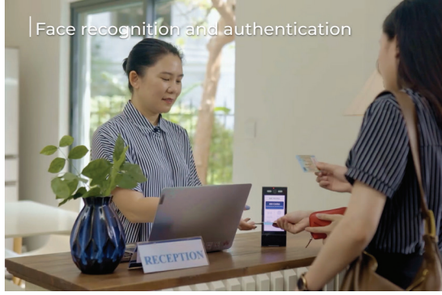
MK Vision's biometric card authentication device provides an identification authentication solution using Match-on Card (MOC) technology to control entry and exit at locations that require human management, such as hotels and business entrances. Data is processed and matched through the chip, reducing the risk of identity forgery and data theft, ensuring accuracy and high reliability for customers using the service.

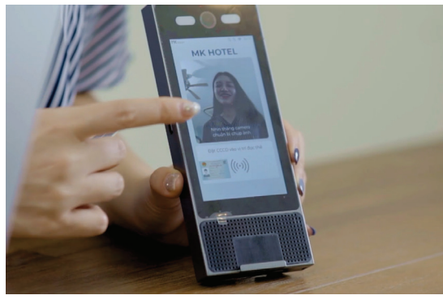
Personnel and Human Resource Management

M6 also integrates with MK Vision's software, offering remote management capabilities, recording all access notifications in real time to help the management center easily grasp and handle situations promptly.

Area Authorization

With the authorization feature, M6 provides solutions in specific areas such as quarantine zones by granting and managing entry permissions for authorized personnel, ensuring synchronized management, access control, and guaranteeing that only authorized individuals can perform access.





Traffic camera solution

Surveillance camera detecting traffic violations

V-ANPR serial

The V-ANPR traffic surveillance camera utilizes edge-based artificial intelligence (AI on Edge) to monitor traffic violations, supporting the processing of fines through automated enforcement; additionally, V-ANPR can measure traffic flow and classify moving vehicles in real-time to promptly issue congestion alerts on the roadway. AI Camera represents an advanced technological solution for an intelligent and sustainable traffic system.

PROCESSOR
ARM Quad-Core 1GHz processor, Separate NPU processing unit

RECOGNITION
- License plates
- Vehicle color
- Vehicle model

DUST/WATER RESISTANCE
IP67 standard

VIOLATION DETECTION
Traffic signal command violation

GPS
GPS & GLONASS / A-GPS

SHOCKPROOF
IK10 standard

SENSOR SIZE
1/2.8"

Violation detection
Driving in exceeded speed limit

WIRELESS CONNECTION
4G LTE

License Plate Recognition

Equipped with license plate recognition technology within a range of 50m-70m (daytime) and 40m-60m (nighttime) with infrared support, the camera's high-resolution image sensor can capture clear images of license plates at speeds ranging from 0-120km/h.

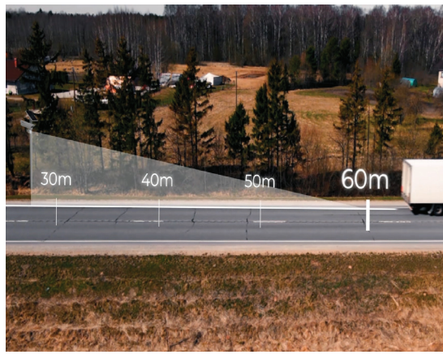
Automatic Violation Detection

The AI processor not only analyzes license plates but also provides information on vehicle type and color. This supports record-keeping for handling violations, issuing alerts, and enhancing traffic safety awareness. The types of violations that V-ANPR can detect include

- Ignoring traffic signals
- Lane encroachment
- Not wearing helmets
- Driving against traffic
- Wrong lane usage
- Speeding (V-Speed)

Real-Time Management

Integrated with MK Vision's traffic monitoring and violation processing software, V-ANPR offers the ability to live-stream video from the camera, recording all notifications in real-time. Additionally, the integration of GIS digital map display and area-wide livestreaming enables the management center to easily grasp and promptly handle occurring situations.



5MP / 8MP

DUAL

Speed measurement camera with recording capability

V-SPEED

V-Speed is designed to monitor vehicle speeds at critical traffic junctions. Leveraging AI and radar modules, V-Speed provides precise detection and alerts for violations, particularly speed violations. It represents the highest configuration model from MK Vision, intended to meet traffic safety monitoring needs on highways with speeds up to 300 km/h.

- **AI on Edge processing**
- **LPR - License Plate Recognition**
- **Intelligent Recording**
- **Speed detection from 8-300km/h**
- **Real-time management and monitoring**
- **Solar Power Source**

5MP

DUAL

Vehicle Speed Detection

The V-SPEED camera is equipped with a speed radar in a range of 8-300 km/h, covering distances of 50-70 meters with fixed installation condition, monitoring in both directions on the lane, and supporting multi-target monitoring. It is capable of operating in low-light conditions on highways with two or more lanes.

Intelligent Violation Processing

With the radar speed violation alerts and LPR technology, the violation processing is both accurate and prompt. When a vehicle is found to be in violation of speed or parking regulations, the edge processing system immediately alerts the monitoring personnel for timely intervention.

Real-time Management

Integrating with MK Vision's software, V-Speed provides direct video monitoring from the camera and records all violation notifications in real-time. Users at the control center can utilize this evidence to issue corresponding penalty notices.

Security Surveillance Camera Solutions

Surveillance IP camera

V-Surveillance serial

V-Surveillance Security Surveillance Camera is a surveillance camera product with a 5MP resolution, supporting security monitoring in all environmental conditions with IP67 dust and water resistance standards. V-Surveillance is designed according to international IP camera standards such as ONVIF and RTSP to easily connect and integrate into a CCTV system comprising multiple interconnected cameras, functioning to monitor designated areas. This allows for expansion across various areas, providing visibility into every corner of the space.

- **ZoneIQ - Intrusion detection and crossing boundary alerts**
- **Open API for software integration**
- **Lenslock - Camera tampering alerts**
- **ONVIF connection protocol**

Enhancing Area Security

With the ZoneIQ feature, V-Surveillance cameras can detect intrusions and provide alerts when boundaries are crossed, enabling timely detection and prevention of unauthorized activities, ensuring the security of the monitored area. Additionally, the device is equipped with Lenslock to detect and alert against camera tampering such as scene changes or abnormal color detection, ensuring the camera always operates normally and is not interfered with or vandalized.

Reliable Evidence Storage

V-Surveillance cameras can continuously record in all weather conditions, capturing clear and detailed images. This enhances the security of data transmission, ensuring that surveillance videos are not lost or illegally accessed, thereby providing reliable evidence when needed.

Software Management Integration Capability

V-Surveillance cameras support the ONVIF connectivity protocol and open API, allowing for easy integration with other security monitoring and management software systems. This creates a synchronized surveillance system and a management center that can promptly handle arising situations.

Security surveillance camera

V-Security Serial

The V-Security surveillance camera system represents an advanced version of the V-Surveillance series, incorporating edge AI processing technology and integrating with security monitoring software to enhance public safety and order in public areas, buildings, and critical infrastructure. V-Security is suited for surveillance scenarios requiring in-depth analysis and proactive alerts for unusual security events.

- **Face recognition up to 20,000 different faces**
- **People counting and crowd detection**
- **Detection of abandoned or missing objects**
- **Define restricted areas and intrusion alerts**
- **Alerts for camera tampering**
- **Intelligent sound analysis**
- **Detection of loitering and wrong-way movement**
- **Heatmap generation**
- **IP68 and IK10 standard**

Comprehensive Security Monitoring

V-Security offers a comprehensive security monitoring solution with the highest quality image capture under various conditions and environments. The product is designed for flexible configuration to meet security needs across different scopes, from narrow to broad, and both day and night.

Edge AI Processing Capabilities

The V-Security system is equipped with edge AI processing capabilities, enabling accurate real-time behavior and object analysis. Optional AI features include:

- [IDVision] Supports up to 20,000 faces in the library with a face recognition rate of ≥90%. Includes face library encryption support.
- [LensLock] Provides tampering alerts (scene change, video blur, and abnormal color detection).
- [ZonelQ] Detects boundary crossings (classifies people/motor vehicles/non-motorized vehicles), and identifies abandoned or lost objects.
- [ZonelQ+] Offers people counting, crowd detection, loitering, heatmap generation, and wrong-way detection.
- [AudiolQ] Analyzes sounds (gunshots, fire alarms, sirens, glass breaking, crying children).

Security Monitoring Software

The system is accompanied by a professional security monitoring software that provides efficient data management, review, and storage capabilities. All alerts and in-depth investigations can be conducted via a single interface, allowing users to easily manage and control the entire monitoring system while optimizing performance.

360 degree security surveillance camera

V-PTZ

V-PTZ security surveillance camera is a high-end monitoring device equipped with remote pan, tilt, and zoom capabilities. With its flexible control features, the PTZ camera allows users to rotate the view angle, tilt the perspective, and adjust the zoom level remotely, enabling detailed observation of a large area. Integrated with AI-based recognition and tracking features, the V-PTZ camera is commonly employed in security surveillance systems in areas such as schools, parks, urban zones, industrial facilities, or enterprises, providing a versatile monitoring solution.

- **360° horizontal field of view, vertical field of view ≥ 90°**
- **Superior performance in low-light conditions**
- **Intrusion tracking and alerting**
- **32x optical zoom, 16x digital zoom**
- **Extended infrared range**
- **People counting and crowd detection**
- **Automatic rotation, recording, and position locking**
- **Tamper alert**
- **Provides heat maps**

Unlimited Observation Capability

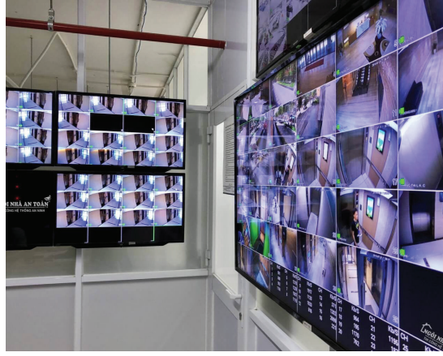
The V-PTZ camera's pan, tilt, and zoom functions allow users to monitor an extensive area without vision limitations. Equipped with a 32x optical zoom, the device has an operational range of up to 800 meters. The PTZ camera's optical capabilities enable users to closely inspect areas with unusual activity, enhancing surveillance and security measures.

AI Edge-Based Tracking Capability

With edge-based AI algorithms, the V-PTZ can intelligently filter objects to focus on human and vehicle detection, suitable for various scenarios to improve surveillance and security efficiency. The device can create a 3D model within its monitoring range, mark, automatically control to track objects upon detection and target selection.

Software Management Integration Capability

The V-PTZ camera supports ONVIF protocol and open API connections, facilitating easy integration with existing surveillance and security management software systems. This function allows for quick system synchronization, cost savings, and flexible expansion.



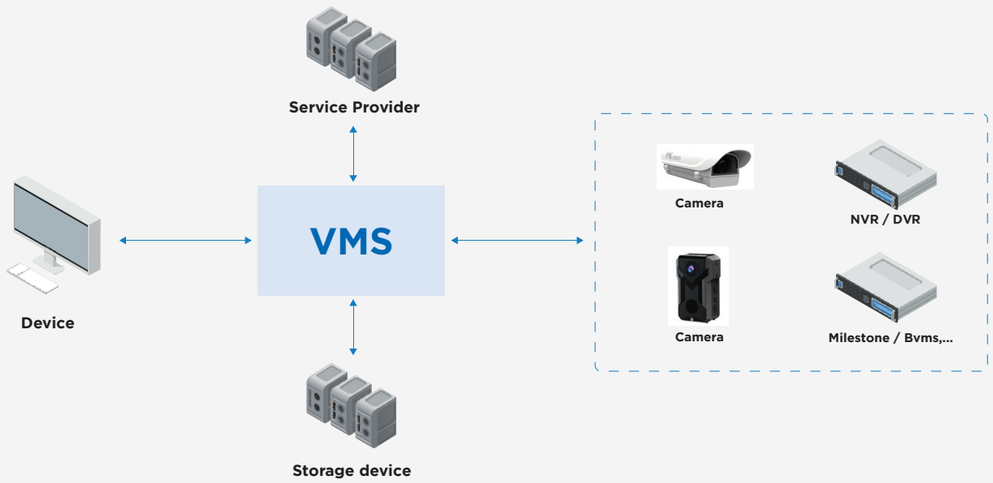
COMS - VMS Software

COMS is an open and specialized management software platform designed for MK Vision's security camera products, featuring professional, easy-to-use, and comprehensive functionalities. More than just a simple video management software, COMS serves as a platform that integrates various security monitoring features such as access control, traffic surveillance, incident handling, team deployment, online communication, and more. COMS is designed with openness, utilizing common integration standards like ONVIF or RTSP, ensuring compatibility with many types of cameras on the market. Data from COMS can also be easily shared through APIs.

- **Manage cameras on a map**
- **Device Management**
- **System Integration with Third Parties**
- **Image Monitoring**
- **User Management**
- **Multi-Platform, Multi-Language Suppor**
- **Event and Violation Management, Violation Classification**
- **Integration of Image Analysis Functions, System Upgrades and Expansion**
- **Dashboard Display for Camera Activity Monitoring, Violation Recognition, and Violation Report Downloads**

Compliance with Ministry of Public Security Standards

Software architecture



COMS

Comprehensive Operating Management System

The software includes internal sub-modules and offers the potential for expanding additional modules in the future.

ITS
Intelligent Transportation System
Intelligent Traffic System and Traffic Flow Management

FMS
Fleet Management System
Fleet Management System (Dashcam)

ACP
Access Control System
Access Control System

IVA
Intelligent Video Analytics
Intelligent Video Analytics System

BWC
Bodyworn camera System
Security Surveillance System

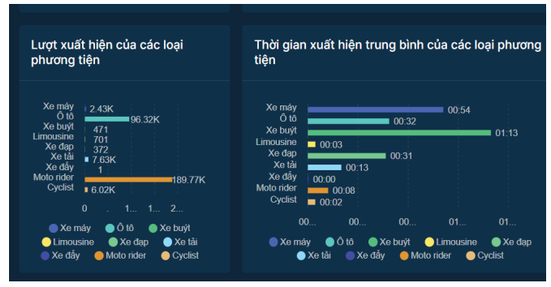
Ai on Edge

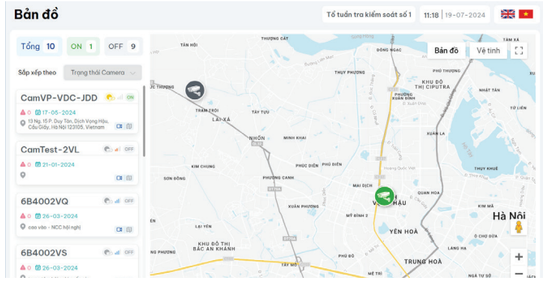
COMS Specification

Workstation configuration	Minimum bandwidth for camera viewing	Maximum number of simultaneous camera views
CPU: Core i3, 10th gen and above SSD: 256GB RAM: 8GB	30 Mbps	9
CPU: Core i5, 10th gen and above SSD: 256GB RAM: 8GB	75Mbps	25
CPU: Core i7, 10th gen and above SSD: 512GB RAM: 16GB GPU: VGA RTX 3060	100Mbps	36
CPU: Core i9, 10th gen and above SSD: 512GB RAM: 32GB GPU: Nvidia Quadro RTX 4000	200Mbps	64

Display Dashboard

- Display camera charts based on status.
- Display and categorize traffic violation charts.
- Allow the download of various violation reports.



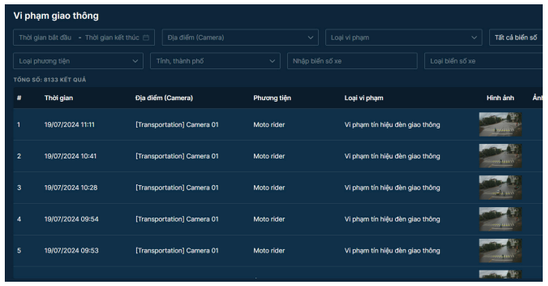


Manage Cameras on the Map

- Display a list of cameras by type.
- Visualize camera locations on the GIS map.
- Display camera operational status on the map.
- Display camera image pop-ups on the map when a warning event occurs.

Manage Events and Violations

- Display a list of events by type.
- View detailed event information.
- Filter by available features.
- Process violations.
- Export violation report lists.



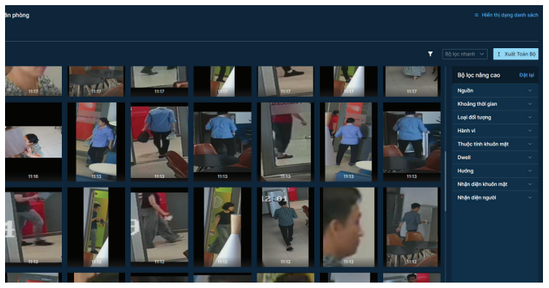
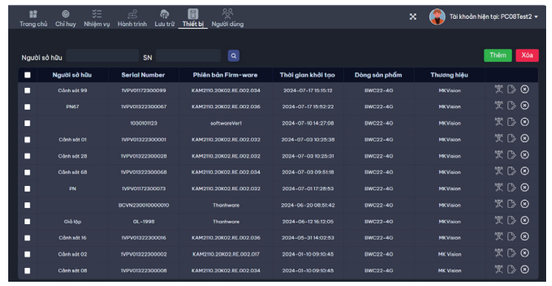


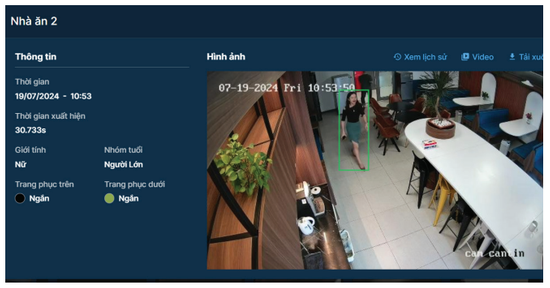
Image Monitoring

- 24/7 monitoring, live viewing in real time.
- Display in multiple modes: 1, 4, 9, 16, 25... cameras on one screen.
- Instant playback.
- Control PTZ cameras.

Device Management

- Display a list of cameras categorized by type.
- Add, edit, delete devices.
- Filter and search for devices by name, SN, camera type, area.
- Configure device parameters.
- Download camera list.





Integrate Image Analysis Functions

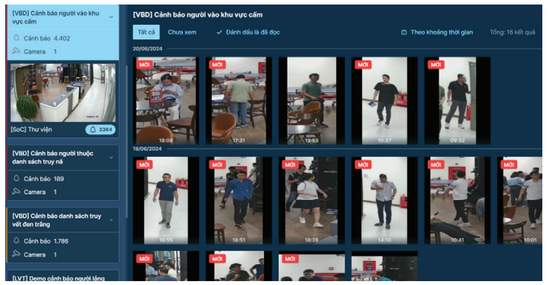
- System is able to upgrade and expand AI features as needed.
- Cameras integrated with AI features such as facial recognition, intrusion detection, crowd warning.
- Compatible with AI View cameras.
- Configure to send alerts when AI events occur.

System Configuration

- Configure cameras.
- Configure maps.
- Configure Dashboard.
- Configure report templates and penalty forms.
- Configure notification settings.
- Configure Web/App settings.
- Configure user management.

Scalability and Upgradability

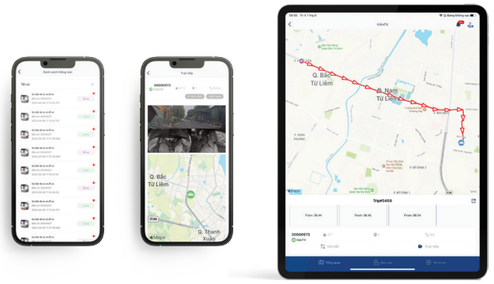
- Flexible deployment model, capable of unlimited camera expansion.
- Allow system configuration upgrades as the number of cameras increases.



TRUNG TÂM NGHIÊN CỨU, ỨNG DỤNG DỮ LIỆU DÂN CƯ VÀ CĂN CƯỚC CÔNG DÂN
RESEARCH AND APPLICATION CENTER FOR RESIDENTS DATA (RAR)

Third-Party Integration

- Pre-integrated with the RAR authentication center.
- Communicate with third-party systems via API.



Other Features

- Support English and Vietnamese languages.
- Support Web browsers.
- Support Mobile App (iOS, Android).

4 TECHNICAL SPECIFICATION

MK vision

AutoSecure

Smart dash camera

Camera	
Sensor	Front of vehicle: Sony 1/2.8" CMOS; Inside cabin: Sony 1/2.8" CMOS
Lens	Front of vehicle: Fixed focal length, 2.8mm, F2.2, 138° viewing angle Inside cabin: Fixed focal length, 1.5mm, F2.8, 190° viewing angle
Night mode	Automatic IR-cut filter Automatic switching
Minimum lighting	Color mode: 0.05 lux, at 50 IRE F2.2 Infrared mode: 0 lux
Configuration	
Processor	Quad-core Arm® Cortex®-A53 1 GHz
Memory	2048 MB DDR4
Processing performance	Ambarella image signal processor and video encoder; CVflow® Edge Vision Analysis
Video	
Resolution	Front: 8MP; Inside cabin: 5MP
Compression standard	H.264
Frame rate	Up to 30 fps at Full HD resolution, Up to 15 fps at 4K resolution
Signal to noise ratio	≥ 50dB
Brightness range (WDR)	≥ 120dB
Adjust the image	Automatic white balance, automatic backlight compensation, automatic gain adjustment
Sound	
Audio receiver	Microphone*1
Audio output	Speaker*1
Audio coding standard	AAC-LC up to 48kHz
Connect	
Wired connection	1*USB-C; 1*Micro-USB (debug)
Wireless Connectivity	Wifi: 802.11b/g/n, 2.4GHz & 5GHz GPS: GPS & GLONASS support A-GPS Mobile data: 4G LTE"
Protocol	MK Vision SDK
Safety information	Digitally signed firmware; digitally sign video; digitally sign events; one-time password (OTP); true random number generator (TRNG); password encryption; TrustZone® secure boot; memory encryption; DRAM shuffle
System integration	
Application programming interface	Open API services for software integration: metadata, events, reports
Video surveillance system	Compatible with MK Vision video management system
Module	DMS camera module ID card reader module (optional)
Analysis	
Smart features	[FleetFocus] Detects 4 speed-related events: Turning, braking, acceler-ation, collision [FleetFocus+] Detects 7 behavior-related events: Drowsiness, level of concentration, phone usage, seat belt/glasses detection, smoke, driver face recognition, liveness recognition [DriverID] Driver identification, anti-fraud
Certification	
ISO	ISO 9001:2015
EMC	FCC, CE Certified
Overview	
Case	ABS plastic, Aluminum
Power supply	11.8 - 36.0V
IR Light	Infrared: 10m
Storage	1024 MB NAND Flash, 1*Micro-SD slot (up to 512 GB)
Operating conditions	-20°C to 65°C, 90% humidity and non-condensing
Storage conditions	-20°C to 70°C, 90% humidity and non-condensing
Size	146 x 93 x 146 mm
Weight	600g
Accessory	Live wire set, Electrostatic film, Glass mounting bracket
Language support	English and Vietnamese
CO	Made in Vietnam
CQ	MK Vision CQ, Quatest

AutoSecure - DAT

Driving school camera solution

Camera	
Sensor	Front of vehicle: Sony 1/2.8" CMOS; Inside cabin: Sony 1/2.8" CMOS
Lens	Front of vehicle: Fixed focal length, 2.8mm, F2.2, 138° viewing angle Inside cabin: Fixed focal length, 1.5mm, F2.8, 190° viewing angle
Night mode	Automatic IR-cut filter; Automatic switching
Minimum lighting	Color mode: 0.05 lux, at 50 IRE F2.2 Infrared mode: 0 lux
Configuration	
Processor	Quad-core Arm® Cortex®-A53 1 GHz
Memory	2048 MB DDR4
Processing performance	Ambarella image signal processor and video encoder; CVflow® Edge Vision Analysis
Video	
Resolution	Front: 8MP; Inside cabin: 5MP
Compression standard	H.264
Frame rate	Up to 30 fps at Full HD resolution; Up to 15 fps at 4K resolution
Signal to noise ratio	≥ 50dB
Brightness range (WDR)	≥ 120dB
Adjust the image	Automatic white balance, automatic backlight compensation, automatic gain adjustment
Sound	
Audio receiver	Microphone*1
Audio output	Speaker*1
Audio coding standard	AAC-LC up to 48kHz
Connect	
Wired connection	1*USB-C; 1*Micro-USB (debug)
Wireless Connectivity	Wifi: 802.11b/g/n, 2.4GHz & 5GHz GPS: GPS & GLONASS support A-GPS Mobile data: 4G LTE
Protocol	MK Vision SDK
Safety information	Digitally signed firmware; digitally sign video; digitally sign events; one-time password (OTP); true random number generator (TRNG); password encryption; TrustZone® secure boot; memory encryption; DRAM shuffle
System integration	
Application programming interface	Open API services for software integration: metadata, events, reports
Video surveillance system	Compatible with MK Vision video management system
Module	DMS camera module ID card reader module with built-in 7 inch touch screen
Analysis	
Smart features	[FleetFocus] Detects 4 speed-related events: Turning, braking, acceler-ation, collision [FleetFocus+] Detects 7 behavior-related events: Drowsiness, level of concentration, phone usage, seat belt/glasses detection, smoke, driver face recognition, liveness recognition [DriverID] Driver identification, anti-fraud [DATM] Display student and teacher information, authentication, display time and distance of practiced students, learner authentication, auto fraudulent activities warning.
Certification	
ISO	ISO 9001:2015
EMC	FCC, CE Certified
Overview	
Case	ABS plastic, Aluminum
Power supply	11.8 - 36.0V
Screen	7 inches touch
IR Light	Infrared: 10m
Storage	1024 MB NAND Flash, 1*Micro-SD slot (up to 512 GB)
Security module	Supports 2 SAM Slots
Operating conditions	-20°C to 65°C, 90% humidity and non-condensing
Storage conditions	-20°C to 70°C, 90% humidity and non-condensing
Size	146 x 93 x 146 mm
Weight	600g
Accessory	Live wire set, Electrostatic film, Glass mounting bracket
Language support	English and Vietnamese
CO	Made in Vietnam
CQ	MK Vision CQ, Quatest

BWC

Body-worn operational camera

Camera	
Sensor	Sony 1/ 2.8" CMOS
Lens	Fixed focal length, 2.8mm, F2.2, 140° viewing angle
Night mode	Automatic conversion
Minimum lighting	Color mode: 0.001 lux, at 50 IRE F2.2 Infrared mode: 0 lux
Configuration	
Processor	Quad-core Arm® Cortex®-A53 1 GHz
Memory	1024MB DDR4
Processing performance	Ambarella image signal processor and video encoder; CVflow® Edge Vision Analysis
Video	
Resolution	2MP
Compression standard	H.264
Frame rate	Up to 30 frames per second at Full HD resolution
Signal to noise ratio	≥ 50dB
Brightness range (WDR)	≥ 120dB
Adjust the image	Automatic white balance, automatic backlight compensation, automatic gain adjustment
Sound	
Audio receiver	Microphone*1
Audio output	Speaker*1
Audio coding standard	AAC-LC up to 48kHz
Connect	
Wired connection	1*USB-C
Wireless Connectivity	Wifi: 802.11b/g/n, 2.4GHz & 5GHz Bluetooth: BLE 4.2 RFID: ISO14443 A/B GPS: GPS & GLONASS support A-GPS Mobile data: 4G LTE
Protocol	MK Vision SDK
Safety information	Digitally signed firmware; digitally sign video; digitally sign events; one-time password (OTP); true random number generator (TRNG); password encryption; TrustZone® secure boot; memory encryption; DRAM shuffle
System integration	
Application programming interface	Open API services for software integration: metadata, events, reports
Video surveillance system	Compatible with MK Vision video management system
Module	
Analysis	
Smart features	[IDVision] Supports up to 20,000 faces in the library, face recognition rate ≥90%; supports face library encryption.
Certification	
ISO	ISO 9001:2015
Protection standard	IP65
EMC	FCC, CE Certified
Overview	
Case	ABS, PC plastic
Power supply	Battery: 4500 mAh Charger: 5V-2A
Screen	2 inches non-touch, 320x240, 600 cd/m2
Storage	64GB EMMC Flash
Security module	Supports 2 SAM Slots
Operating conditions	-10°C to 45°C, 90% humidity and non-condensing
Storage conditions	-10°C to 50°C, 90% humidity and non-condensing
Size	101 x 64 x 31 mm
Weight	205 g
Accessory	Charging base, Clothes clip, USB-C cable, 5V-2A charger
Language support	English and Vietnamese
CO	Made in Vietnam
CQ	MK Vision CQ, Quatest

V-Surveillance Serial

Surveillance IP camera

Camera	
Version	**2MP / 5MP / 8MP**
Sensor	**2MP / 5MP:** Sony 1/2.5" CMOS **8MP:** Sony 1/2.8" CMOS
Lens	Auto-adjustable focal length, 2.8 - 12mm, F1.6, 129° viewing angle Automatically adjusts lens aperture.
Night mode	Automatic IR-cut filter Automatic switching
Minimum lighting	Color mode: 0.05 lux, at 50 IRE F2.2 Infrared mode: 0 lux
Configuration	
Processor	Quad-core Arm® Cortex®-A53 1 GHz
Memory	1024 MB DDR4
Processing performance	Ambarella image signal processor and video encoder; CVflow® Edge Vision Analysis
Video	
Resolution	2MP / 5MP / 8MP
Compression standard	H.265 / H.264 / Motion JPEG
Frame rate	Up to 30 frames per second at Full HD resolution
Signal to noise ratio	≥ 50dB
Brightness range (WDR)	≥ 120dB
Adjust the image	Automatic white balance, automatic backlight compensation, automatic brightness gain adjustment
Sound	
Audio receiver	Microphone*1
Audio coding standard	AAC-LC 48kHz
Connect	
Wired connection	1*RJ45
Wireless Connectivity	**2MP & 5MP:** Wifi: 802.11 b/g/n, 2.4GHz & 5GHz GPS: GPS & GLONASS support A-GPS **8MP:** Wifi: 802.11 b/g/n, 2.4GHz & 5GHz GPS: GPS & GLONASS support A-GPS Mobile data: 4G LTE (optional)
Protocol	ONVIF profile G,S,T
Safety information	Digitally signed firmware; digitally sign video; digitally sign events; one-time password (OTP); true random number generator (TRNG); password encryption; TrustZone® secure boot; memory encryption; shuffle DRAM; IP address filtering; video encryption; firmware encryption; configuration encryption; unauthorized login lockout; IP address filtering; 802.1x authentication; basic authentication; digest authentication for HTTP
System integration	
Application programming interface	Software integration API: metadata, events, reports
Video surveillance system	Compatible with MK Vision video management system
Analysis	
Smart features	[LensLock] Supports camera interference warnings (scene change, video blur, and abnormal color detection). [ZoneIQ] Supports lines/zones crossing detection for people and motor/ non-motorized vehicles, abandoned/lost objects detection
Certification	
ISO	ISO 9001:2015
Protection standard	IP67, IK10
EMC	FCC, CE Certified
Overview	
Case	Aluminum
Power supply	12VDC; PoE IEEE 802.3af
IR Light	Infrared: ≥ 50m
Storage	64GB EMMC Flash
Security module	Supports 2 SAM Slots
Operating conditions	-10°C to 60°C, 95% humidity and non-condensing
Storage conditions	-10°C to 65°C, 95% humidity and non-condensing
Size	218 x 81 x 80 mm
Weight	680g
Accessory	Hanging brackets, screws
Language support	English and Vietnamese
CO	Made in Vietnam
CQ	MK Vision CQ, Quatest

V-PTZ

360 degree security surveillance camera

Camera	
Version	**5MP / 8MP**
Sensor	5MP: Sony 1/2.5" CMOS 8MP: Sony 1/2.8" CMOS
Lens	Auto-adjustable focal length, 4.8 - 154mm, 32x optical zoom, 16x digital zoom Automatically adjusts lens aperture
Night mode	Automatic IR-cut filter Automatic switching
Minimum lighting	Color mode: 0.01 Lux @ (F1.3,AGC ON) Black and white mode: 0.002 Lux @(F1.3 AGC ON) Infrared mode: 0 Lux
PTZ mode	Maximum rotation angle horizontally 360°, vertically ≥ 90° Rotation speed 0.1°-210°/s Automatic rotation setting, random rotation Automatically remembers position when power off
Video	
Resolution	**5MP / 8MP**
Compression standard	H.265 / H.264 / Motion JPEG
Frame rate	Up to 30 frames per second at Full HD resolution
Signal to noise ratio	≥ 50dB
Brightness range (WDR)	≥ 120dB
Adjust the image	Automatic white balance, automatic backlight compensation, automatic brightness gain adjustment
Sound	
Audio receiver	5MP: Microphone*1 8MP: 1 audio input; 1 audio output
Connect	
Wired connection	1*RJ45
Protocol	ONVIF profile G,S,T
Safety information	5MP: Video encryption; firmware encryption; configuration encryption; unauthorized login blocking; IP address filtering; 802.1x authentication; basic authentication; digest authentication for HTTP 8MP: User authentication (ID and PW); server authentication (MAC address); HTTPS encryption; network access based on IEEE802.1x port control; filter IP addresses
System integration	
Application programming interface	Software integration API: metadata, events, reports
Video surveillance system	Compatible with MK Vision video management system
Analysis	
Smart features	[LensLock] Supports camera interference warnings (scene change, video blur, and abnormal color detection). [ZoneIQ] Supports lines/zones crossing detection for people and motor/non-motorized vehicles, abandoned/lost objects detection
Certification	
ISO	ISO 9001:2015
Protection standard	IP67, IK10
EMC	FCC, CE Certified
Overview	
Case	Aluminum
Power supply	12VDC/ 24 VDC; PoE IEEE 802.3af
IR Light	Infrared: 200m
Storage	1*Micro-SD slot (up to 256 GB)
Operating conditions	-45°C to 65°C, 90% humidity and non-condensing
Storage conditions	-40°C to 70°C, 95% humidity and non-condensing
Accessory	Hanging brackets, screws
Language support	English and Vietnamese
CO	Made in Vietnam
CQ	MK Vision CQ, Quatest

M6

E-ID and biometric reader

Camera	
Sensor	Sony 1/ 2.8" CMOS
Lens	Fixed focal length, 2.8mm, F2.2, 140° viewing angle
Night mode	Automatic conversion
Minimum lighting	Color mode: 0.001 lux, at 50 IRE F2.2 Infrared mode: 0 lux
Configuration	
Processor	Quad-core Arm® Cortex®-A53 1 GHz
Memory	1025MB DDR4
Processing performance	Ambarella image signal processor and video encoder; CVflow® Edge Vision Analysis
Video	
Resolution	2MP
Compression standard	H.264
Frame rate	Up to 30 frames per second at Full HD resolution
Signal to noise ratio	≥ 50dB
Brightness range (WDR)	≥ 120dB
Adjust the image	Auto white balance, auto backlight compensation, auto gain adjustment, HDR
Sound	
Audio receiver	Microphone*1
Audio output	Speaker*1
Audio coding standard	AAC-LC up to 48kHz
Connect	
Wired connection	1*USB-C, 1*Relay port, 1*Alarm, 1*RJ45
Wireless Connectivity	Wifi: 802.11b/g/n, 2.4GHz & 5GHz Bluetooth: BLE 4.2 NFC: Support for reading ISO/IEC 14443 contactless protocol cards.
Protocol	MK Vision SDK
Safety information	Digitally signed firmware; digitally sign video; digitally sign events; one-time password (OTP); true random number generator (TRNG); password encryption; TrustZone® secure boot; memory encryption; DRAM shuffle
System integration	
Application programming interface	Open API services for software integration: metadata, events, reports
Video surveillance system	Compatible with MK Vision video management system
Analysis	
Smart features	[IDVision] Supports up to 20,000 faces in the library; face recognition rate ≥90%, supports face library encryption, supports fingerprint recognition with the FAP20 optical fingerprint module, compatible with fingerprint data in chip-embedded e-ID, automatically detects live fingerprints.
Certification	
ISO	ISO 9001:2016
EMC	FCC, CE Certified
Overview	
Case	ABS, PC plastic
Power supply	12VDC; PoE IEEE 802.3af
Screen	5 inches touch, 480x854 px
IR Light	White LED: 0.5m
Storage	64GB EMMC Flash
Security module	Supports 2 SAM Slots
Operating conditions	-10°C to 50°C, 90% humidity and non-condensing
Storage conditions	-10°C to 50°C, 90% humidity and non-condensing
Size	185 x 79 x 27 mm
Weight	130 g
Accessory	Tabletop base, Wall mount, Relay cable, Alarm cable
Language support	English and Vietnamese
CO	Made in Vietnam
CQ	MK Vision CQ, Quatest

V-SPEED Serial

Speed measurement camera with recording capability

5MP

DUAL

Camera	
Version	**5MP / DUAL**
Sensor	Sony 2/3" Global shutter CMOS
Lens	Auto-adjustable focus, 10-40mm, F1.3, 45° viewing angle 2-lane monitoring capability
Night mode	Automatic IR-cut filter; Automatic switching
Minimum lighting	Color mode: 0.05 lux, at 50 IRE F1.3; Infrared mode: 0 lux
Radar	
Speed measurement range	-320km/h - +320km/h
Speed error	-2km/h - +2km/h
Measurement mode	Fixed, reverse, forward, day and night
Measuring technology	Capable of tracking multiple targets
Monitoring capacity	2-6 lanes
Influenced by cosine angle effect	No
Configuration	
Processor	Ambarella CV25 Quad-core Arm® Cortex®-A53 1 GHz
Memory	2048 MB DDR4
Processing performance	Ambarella image signal processor and video encoder; CVflow® Edge Vision Analysis
Video	
Resolution	5MP
Compression standard	H.265/H.264/MJPEG
Frame rate	Up to 30 frames per second at Full HD resolution
Signal to noise ratio	≥ 50dB
Brightness range (WDR)	≥ 120dB
Adjust the image	Automatic white balance, automatic backlight compensation, automatic gain adjustment, automatic light correction
Analysis	
Smart features	[LensLock] Supports camera vandalism warnings (scene changes, video blur, and abnormal color detection) [ANPR] Supports license plate recognition for motorbikes and cars at a maximum speed of 180 km/h, license plate recognition rate ≥ 95% during the day, ≥ 80% at night. [CarID] Supports vehicle attribute identification, including vehicle type, color, brand (4800 vehicle models included). [TrafficIQ] Supports vehicle direction detection, axle counting, speed estimation, traffic measurement.
Connect	
Wired connection	1*USB-C; 1*Micro-USB (debug), 1*RJ45
Wireless Connectivity	Wifi: 802.11b/g/n, 2.4GHz & 5GHz; GPS: GPS & GLONASS support A-GPS Mobile data: 4G LTE (optional)
Protocol	ONVIF profiles G,S,T
Safety information	Digitally signed firmware; digitally sign video; digitally sign events; one-time password (OTP); true random number generator (TRNG); password encryption; TrustZone® secure boot; memory encryption; shuffle DRAM; IP address filtering; video encryption; firmware encryption; configuration encryption; unauthorized login lockout; IP address filtering; 802.1x authentication; basic authentication; digest authentication for HTTP
System integration	
Application programming interface	Open API services for software integration: metadata, events, reports
Video surveillance system	Compatible with MK Vision video management system
Certification	
ISO	ISO 9001:2015
Protection standard	IP68, IK10
EMC	FCC, CE Certified
Overview	
Case	Aluminum
Power supply	11.8 - 36.0VDC; PoE IEEE 802.3af
IR Light	Infrared: 50-140m
Storage	1024 MB NAND Flash, 1*Micro-SD slot (up to 512 GB)
Security module	Supports 2 SAM Slots
Operating conditions	-20°C to 65°C, 90% humidity and non-condensing
Storage conditions	-20°C to 70°C, 90% humidity and non-condensing
Size	503 x 170 x 159 mm
Weight	3200g
Accessory	Hanging brackets, screws
Language support	English and Vietnamese
CO	Made in Vietnam
CQ	MK Vision CQ, Quatest

V-ANPR

Surveillance camera detecting traffic violations

5MP/ 8MP

DUAL

Camera	
Version	**5MP / 8MP / DUAL**
Sensor	Sony 2/3" Global shutter CMOS
Lens	10-40mm, F1.3, 2-lane monitoring capability 2.8-8mm, F1.3, 4-lane monitoring capability Automatically adjusts light
Night mode	Automatic IR-cut filter Automatic switching
Minimum lighting	Color mode: 0.05 lux, at 50 IRE F1.3 Infrared mode: 0 lux
Configuration	
Processor	Ambarella CV25 Quad-core Arm® Cortex®-A53 1 GHz
Memory	1024 MB DDR4
Processing performance	Ambarella image signal processor and video encoder; CVflow® Edge Vision Analysis
Video	
Resolution	5MP 8MP
Compression standard	H.265/H.264/MJPEG
Frame rate	Up to 30 frames per second at Full HD resolution
Signal to noise ratio	≥ 50dB
Brightness range (WDR)	≥ 120dB
Adjust the image	Automatic white balance, automatic backlight compensation, automatic brightness gain adjustment,
Connect	
Wired connection	1*USB-C; 1*Micro-USB (debug), 1*RJ45
Wireless Connectivity	Wifi: 802.11 b/g/n, 2.4GHz & 5GHz GPS: GPS & GLONASS support A-GPS Mobile data: 4G LTE (optional)
Protocol	ONVIF profiles G,S,T
Safety information	Digitally signed firmware; digitally sign video; digitally sign events; one-time password (OTP); true random number generator (TRNG); password encryption; TrustZone® secure boot; memory encryption; shuffle DRAM; IP address filtering, video encryption; firmware encryption; configuration encryption; unauthorized login lockout; IP address filtering; 802.1x authentication; basic authentication; digest authentication for HTTP
System integration	
Application programming interface	Open API services for software integration: metadata, events, reports
Video surveillance system	Compatible with MK Vision video management system
Analysis	
Smart features	[LensLock] Supports camera vandalism warnings (scene changes, video blur, and abnormal color detection) [ANPR] Supports license plate recognition for motorbikes and cars at a maximum speed of 180 km/h, license plate recognition rate ≥ 95% during the day, ≥ 80% at night. [CarID] Supports vehicle attribute recognition, including vehicle type, color, brand (4800 video samples included). [TrafficIQ] Supports vehicle direction detection, axle counting, speed estimation, supports individual traffic violation detection: lane violation, reckless overtaking, illegal park/stop, usage of prohibited vehicles on highway, road direction violation, traffic signal violation, traffic flow analytics.
Certification	
ISO	ISO 9001:2015
Protection standard	IP68, IK10
EMC	FCC, CE Certified
Overview	
Case	Aluminum
Power supply	11.8 - 36.0VDC; PoE IEEE 802.3af
IR Light	Infrared: 50-140m
Storage	1024 MB NAND Flash, 1*Micro-SD slot (up to 512 GB)
Security module	Supports 2 SAM Slots
Operating conditions	-20°C to 65°C, 90% humidity and non-condensing
Storage conditions	-20°C to 70°C, 90% humidity and non-condensing
Size	503 x 170 x 159 mm
Weight	3200g
Accessory	Hanging brackets, screws
Language support	English and Vietnamese
CO	Made in Vietnam
CQ	MK Vision CQ, Quatest

V-Security

Security surveillance camera

Camera	
Version	**5MP / 8MP**
Sensor	5MP: Sony 1/2.5" CMOS 8MP: Sony 1/2.8" CMOS
Lens	Auto-adjustable focal length, 2.8 - 12mm, F1.6, 129° viewing angle Automatically adjusts lens aperture.
Night mode	Automatic IR-cut filter Automatic switching
Minimum lighting	Color mode: 0.05 lux, at 50 IRE F2.2 Infrared mode: 0 lux
Configuration	
Processor	Ambarella CV25 Quad-core Arm® Cortex®-A53 1 GHz
Memory	5MP: 1024 MB DDR4 8MP: 2048 MB DDR4
Processing performance	Ambarella image signal processor and video encoder; CVflow® Edge Vision Analysis
Video	
Resolution	5MP / 8MP
Compression standard	H.265 / H.264 / Motion JPEG
Frame rate	Up to 30 frames per second at Full HD resolution
Signal to noise ratio	≥ 50dB
Brightness range (WDR)	≥ 120dB
Adjust the image	Automatic white balance, automatic backlight compensation, automatic bright-ness gain adjustment
Sound	
Audio receiver	Microphone*1
Audio coding standard	AAC-LC 48kHz
Connect	
Wired connection	1*RJ45
Wireless Connectivity	Wifi: 802.11 b/g/n, 2.4GHz & 5GHz GPS: GPS & GLONASS support A-GPS [8MP] Mobile data: 4G LTE (optional)
Protocol	ONVIF profile G,S,T
Safety information	Digitally signed firmware; digitally sign video; digitally sign events; one-time password (OTP); true random number generator (TRNG); password encryption; trustzone® secure boot; memory encryption; shuffle DRAM; IP address filtering; video encryption; firmware encryption; configuration encryption; unauthorized login lockout; IP address filtering; 802.1x authentication, basic authentication; digest authentication for HTTP
System integration	
Application programming interface	Software integration API: metadata, events, reports
Video surveillance system	Compatible with MK Vision video management system
Analysis	
Smart features	[IDVision] Supports up to 20,000 faces in the library, face recognition rate ≥90%, supports face library encryption [LensLock] Supports camera interference warnings (scene change, video blur, and abnormal color detection) [ZonelQ] Supports lines/zones crossing detection for people and motor/non-motorized vehicles, abandoned/lost objects detection [ZonelQ+] Counting people, detecting crowds, loitering, heat map, going in the wrong direction [AudiolQ] Supports audio analysis (gun shots, fire alarms, sirens, glass breaking, children crying)
Certification	
ISO	ISO 9001:2015
Protection standard	IP67, IK10
EMC	FCC, CE Certified
Overview	
Case	Aluminum
Power supply	5MP: 12VDC; PoE IEEE 802.3af 8MP: 11.8 - 36.0VDC; PoE IEEE 802.3af
IR Light	≥ 50m
Storage	5MP: 64 GB EMMC Flash 8MP: 1024 MB NAND Flash, 1*Micro-SD slot (up to 512 GB)
Security module	Supports 2 SAM Slot
Operating conditions	5MP: -10°C to 60°C, 95% humidity and non-condensing 8MP: -20°C to 65°C, 90% humidity and non-condensing
Storage conditions	5MP: -10°C to 65°C, 95% humidity and non-condensing 8MP: -20°C to 70°C, 90% humidity and non-condensing
Size	5MP: 218 x 81 x 80 mm 8MP: 503 x 170 x 159 mm
Weight	5MP: 680g 8MP: 3200g
Accessory	Hanging brackets, screws
Language support	English and Vietnamese
CO	Made in Vietnam
CQ	MK Vision CQ, Quatest

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Online documents

MK vision

MK Vision's AI Camera factory

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